

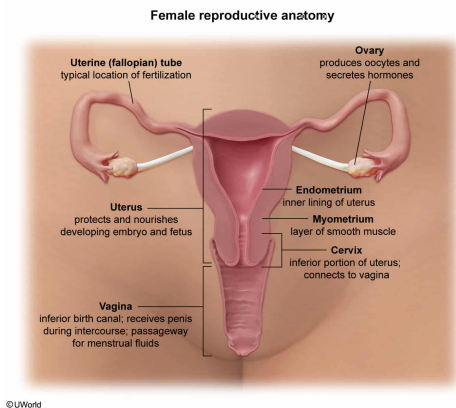
Which statement describes an anatomical characteristic of the uterus?

- ☐ A. The uterus opens into the vagina at the uterine tube. [21%]
- ✓ ☒ B. The uterus includes a layer called myometrium. [52%]
- ☐ C. The uterus contains a thick layer of striated muscle. [16%]
- ☐ D. The uterus is generally where fertilization occurs. [9%]

Correct

53% answered correctly

Explanation:



The female reproductive system is responsible for producing female **gametes** (ie, ova) and protecting, nourishing, and providing support to a developing embryo and fetus. The system includes the following internal structures:

- The **ovaries** are reproductive organs that **produce oocytes** (immature gametes) via oogenesis and secrete **hormones** (eg, **estrogens**, progesterone, inhibin).
- The **uterine tubes** (fallopian tubes, oviducts) are tubes connecting each ovary to the uterus and are the typical location of **fertilization (Choice D)**.
- The **uterus** is muscular organ responsible for protecting and nourishing the embryo and fetus. The uterus is lined with an inner layer called the **endometrium**, which changes in thickness at different points in the monthly uterine cycle. The uterus also includes a **thick** layer of **smooth muscle** (ie, **myometrium**) involved in contractions during childbirth.
 - The **cervix** is the most inferior portion of the uterus and serves as the opening into the vagina (**Choice A**). During childbirth, the cervix thins and the fetus passes through the cervix into the vagina.
- The **vagina** is a muscular tube that functions in the elimination of menstrual fluids during the **menstrual cycle**, the reception of the male penis during sexual intercourse, and as the final segment of the birth canal during delivery.

(Choice C) The type of muscle the uterus contains is smooth muscle that *lacks* striations, unlike skeletal and cardiac muscle.

Educational objective:

As part of the female reproductive system, the uterus nourishes the embryo and fetus. It is lined with endometrium, includes a layer of smooth muscle called myometrium, and connects to the vagina at the cervix.

Time spent: 0 sec
Subject:
Biology

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System:
Reproduction